

Sample End Semester Question Paper - 2025

Subject: Computer Organization and Architecture | Code: CS501 | Seme

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define a page table.
2. What is a pipeline?
3. Name two types of pipeline hazards.
4. What is SIMD?
5. State one difference between RISC and CISC.
6. What is the von Neumann architecture?
7. Name two CPU registers.
8. What is an instruction cycle?
9. Name two I/O methods.
10. What is cache memory?

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain cache memory and mapping techniques with an example.
2. Explain virtual memory and page tables.
3. Explain pipelining: types of hazards and their solutions.
4. Compare RISC and CISC architectures. Explain parallelism (SIMD/MIMD).
5. Explain the von Neumann architecture and its limitations.
6. Explain the CPU: ALU, registers and control unit with a block diagram.
7. Explain the instruction cycle, timing and addressing modes.
8. Explain I/O methods: programmed I/O, interrupt-driven I/O and DMA.